EXPERIMENT 06

Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging

Library Book Issue Management — Workflow Automation Use Case

Sreeman Naibal S

241401106

1. Aim

To automate a key business process of the Library Book Issue Management application — sending a book issue request confirmation email the moment a member submits the library management form — using a Zoho Creator workflow and a Deluge script, and to test and debug the automation end-to-end with real form submissions.

This removes the need for library staff to manually notify every member, ensuring each submitted book issue request is acknowledged immediately and consistently.

2. Procedure

Step 1: Start a new workflow — From the application's Workflow section, a new workflow was created on the library management form.

Step 2: Define the trigger — The Record Event was set to Created and the Form Event (When to trigger workflow) was set to Successful form submission, so the automation fires only after a valid, saved submission.

Step 3: Name the workflow — The workflow was named on successful form submission and saved using Create Workflow.

Step 4: Add a new action — In the Add New Action dialog, Deluge script was chosen as the Action type so custom logic could compose and send the email.

Step 5: Write the Deluge script — The form inputs Email, Name, Book_category, Preferred_Issue_Date and Preferred_Collection_Time were first stored in the variables member_email, member_name, book_category, issue_date and collection_time (the date and time converted with toString()). A sendmail script then sets from to zoho.adminuserid, to to member_email, builds a Subject of “Library Book Issue Request Submitted”, and concatenates a message with the member's name, book category, preferred issue date and preferred collection time.

Step 6: Save the action — The script was saved so that it executes automatically under library management › Created › Successful form submission.

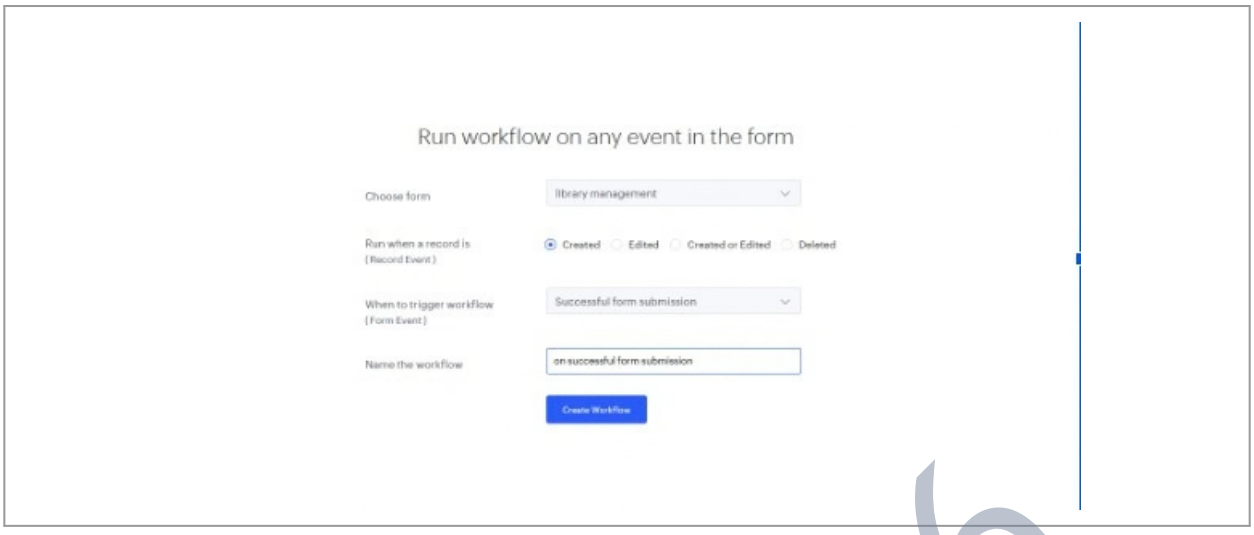
Step 7: Test with a live submission — The library management form was opened in the live application and filled in through its Member details and Contact details sections — Book Category: Computer Science & Engineering, Preferred Issue Date: 29-Oct-2026 — and submitted to trigger the workflow.

Step 8: Debug and verify the output — The member's inbox was checked immediately after submission to confirm the automation ran correctly: the confirmation email arrived with the correct dynamic values (name, book category, issue date and collection time), confirming the Deluge script executed without errors.

241401106

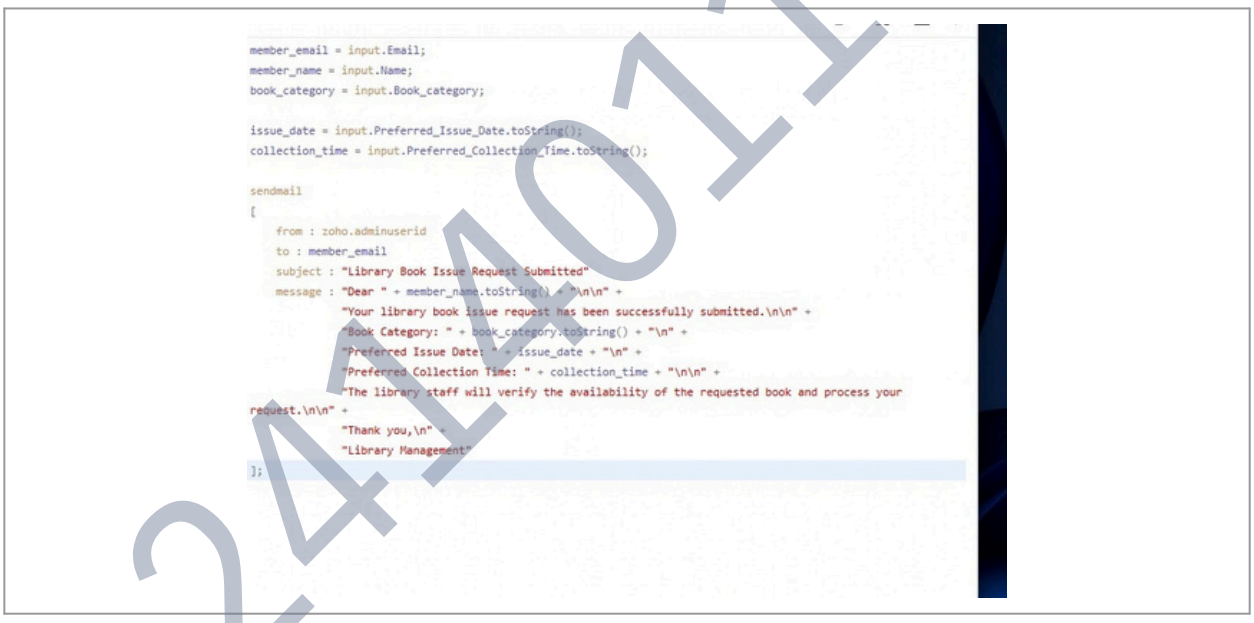
3. Output

3.1 Creating the workflow — trigger configuration



The new-workflow screen shows the form set to library management, the Record Event set to Created, and the Form Event set to Successful form submission, with the workflow named on successful form submission. This defines exactly when the automation should run.

3.2 Writing the Deluge script



```
member_email = input.Email;
member_name = input.Name;
book_category = input.Book_category;

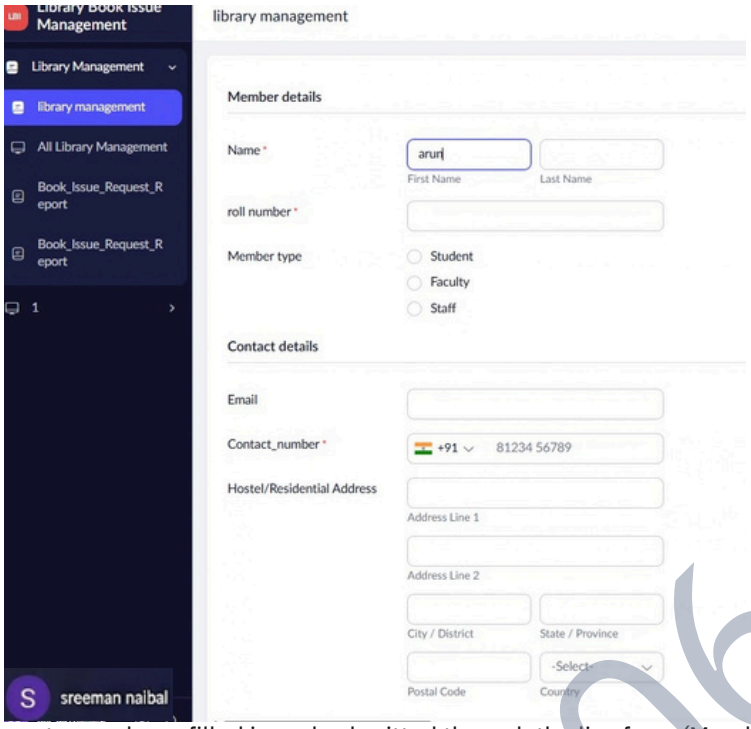
issue_date = input.Preferred_Issue_Date.toString();
collection_time = input.Preferred_Collection_Time.toString();

sendmail
[
    from : zoho.adminuserid
    to : member_email
    subject : "Library Book Issue Request Submitted"
    message : "Dear " + member_name.toString() + "\n\n" +
              "Your library book issue request has been successfully submitted.\n\n" +
              "Book Category: " + book_category.toString() + "\n" +
              "Preferred Issue Date: " + issue_date + "\n" +
              "Preferred Collection Time: " + collection_time + "\n\n" +
              "The library staff will verify the availability of the requested book and process your
request.\n\n" +
              "Thank you,\n" +
              "Library Management"
];
```

The Deluge editor shows the completed script. It reads the form fields Email, Name, Book_category, Preferred_Issue_Date and Preferred_Collection_Time into variables, then uses sendmail to build the subject and message dynamically and send them to member_email.

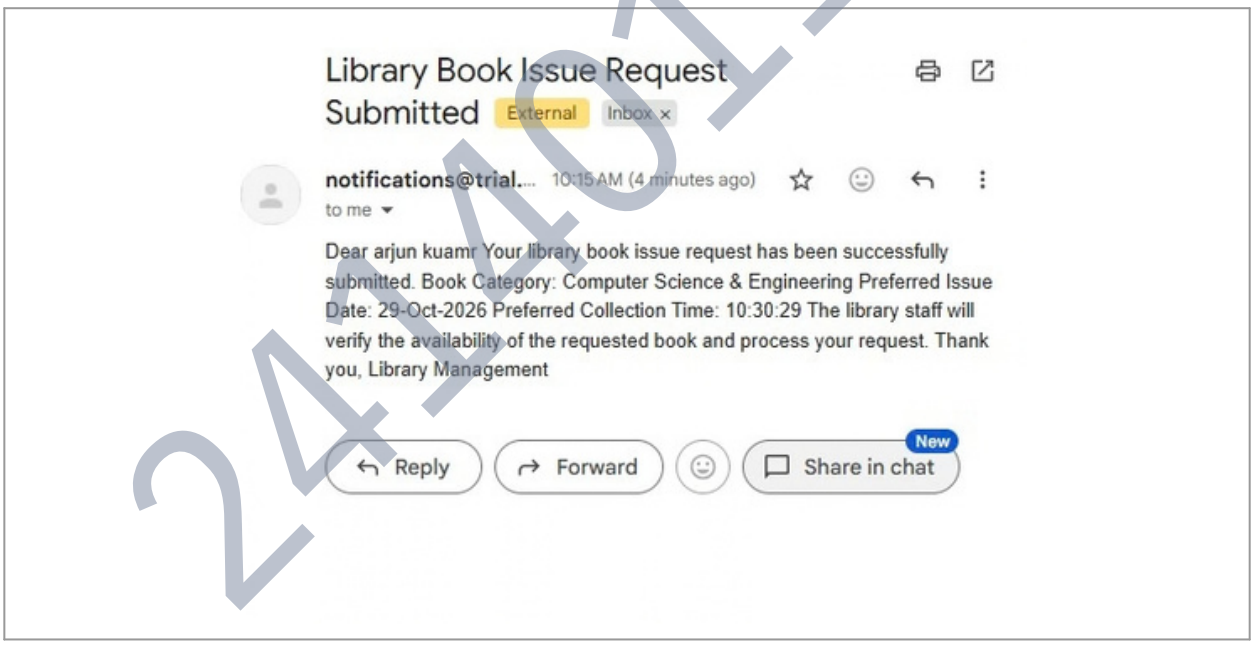
24101106

3.3 Testing — submitting a live record



A real library management record was filled in and submitted through the live form (Member details, Contact details, Book Category: Computer Science & Engineering, Preferred Issue Date 29-Oct-2026) specifically to trigger the on successful form submission workflow and observe its behaviour.

3.4 Debugging output — confirmation email received



The received email confirms the automation ran successfully: the subject and message match the Deluge script exactly, and every dynamic value (Member Name, Book Category: Computer Science & Engineering, Preferred Issue Date: 29-Oct-2026, Preferred Collection Time: 10:30:29) was pulled correctly from the submitted record, verifying the workflow end-to-end.

24140106

4. Final Result

A complete workflow was built on the library management form that fires automatically on successful submission, using a Deluge sendmail script to generate a personalised confirmation email with no manual intervention from the library staff.

The automation was tested with a live form submission and debugged by inspecting the resulting email: all dynamic fields rendered correctly, confirming the script's field references and string concatenation were error-free.

The exercise met its objective: a real business process — notifying members that their book issue request was received — is now fully automated through low-code workflow and Deluge scripting, and the testing step confirmed the automation behaves reliably end-to-end.

24140106